

PATENSIE, South Africa

WMO: 688350

Lat: 33.7653S

Lon: 24.8233E

Elev: 318

StdP: 14.53

Time Zone: 2.00 (E02)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	40.3	42.6	27.4	21.6	66.3	30.8	25.3	63.9	18.4	69.4	16.2	68.9	1.9	340	0.300

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	17.5	91.6	69.7	87.9	69.2	85.1	68.2	73.4	83.4	72.2	82.1	71.0	80.5	6.9	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.7	114.6	76.3	69.4	109.2	75.8	68.1	104.3	74.7	37.2	83.0	36.1	83.0	35.0	80.3	77.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.7	12.1	10.1	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	65.4	72.6	73.5	71.1	66.2	63.1	58.3	58.1	58.4	61.0	64.5	67.4	71.6		
	DBStd	7.44	4.48	4.29	4.89	5.57	4.84	5.18	5.61	4.67	4.66	5.16	5.00	4.83		
	HDD50	5	0	0	0	0	0	1	3	1	0	0	0	0		
	HDD65	1049	2	1	6	49	97	212	228	212	137	71	30	4		
	CDD50	5642	701	659	654	485	406	251	253	260	330	450	523	670		
	CDD65	1213	238	239	196	84	37	13	14	7	17	56	103	209		
	CDH74	7529	1423	1359	1067	481	275	138	158	92	142	400	605	1389		
	CDH80	2276	444	435	314	156	63	23	26	15	49	134	153	464		
(14) Wind	WSAvg	3.3	3.4	3.2	2.8	2.6	2.8	3.6	3.8	3.3	3.5	3.4	3.8	3.6		
(15) Precipitation	PrecAvg	22.7	1.7	1.7	2.4	2.0	1.3	1.4	1.9	2.2	1.7	2.0	2.1	1.8		
	PrecMax	35.7	3.9	4.4	6.6	7.2	4.8	4.7	4.7	7.9	5.3	5.6	5.2	7.6		
	PrecMin	16.9	0.4	0.6	0.5	0.6	0.2	0.3	0.3	0.4	0.3	0.7	0.4	0.3		
	PrecStd	4.9	1.0	1.1	1.6	1.6	1.1	1.1	1.5	2.0	1.4	1.3	1.4	1.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	94.1	94.0	92.7	92.5	85.9	83.2	83.2	82.2	89.1	92.8	89.7	95.0		
		MCWB	71.1	71.8	70.3	66.8	61.2	57.9	57.7	58.4	61.8	66.7	68.0	70.8		
	2%	DB	88.6	88.7	87.2	83.9	80.5	78.1	78.2	76.1	77.1	82.8	84.6	88.8		
		MCWB	71.3	71.3	69.3	65.2	60.9	55.7	56.2	57.4	60.3	65.6	67.3	70.0		
	5%	DB	85.2	85.0	83.3	79.0	76.2	73.6	73.7	71.5	73.1	77.8	80.7	84.6		
		MCWB	69.7	70.1	68.4	64.6	60.5	55.0	55.7	57.1	59.6	63.5	65.6	68.9		
	10%	DB	82.0	81.9	80.1	75.1	72.2	69.0	69.0	68.0	69.7	73.8	77.2	81.1		
		MCWB	68.3	69.0	67.5	63.0	59.8	54.2	54.8	55.8	58.6	62.0	64.3	67.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	75.0	74.6	73.7	70.0	66.3	61.6	61.1	64.0	65.6	70.3	71.1	74.1		
		MCDB	85.9	86.2	82.1	81.8	74.6	72.1	71.3	74.5	77.1	84.8	84.4	86.2		
	2%	WB	73.2	73.1	72.4	67.7	63.9	59.3	59.4	61.1	63.1	67.1	68.9	72.4		
		MCDB	82.8	83.5	81.0	80.0	71.9	68.7	70.1	68.3	73.1	78.0	80.0	83.3		
	5%	WB	71.6	71.8	70.7	66.1	62.4	57.6	58.0	59.3	61.3	65.1	67.3	70.7		
		MCDB	81.4	81.1	79.3	76.2	71.2	67.9	68.6	65.7	70.3	74.4	76.7	81.1		
	10%	WB	70.1	70.6	69.2	64.6	61.1	56.1	56.7	57.5	59.4	63.4	65.8	69.2		
		MCDB	79.4	79.0	77.4	72.9	69.5	65.8	66.5	64.7	67.9	71.8	74.7	78.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	17.6	17.5	17.8	18.5	20.5	21.5	20.7	20.7	18.9	18.0	17.7	18.5		
		MCDBR	23.2	23.4	23.8	25.1	26.2	27.6	27.7	26.4	24.8	26.4	24.8	25.7		
	5% WB	MCWBR	9.0	8.8	8.8	10.0	10.4	11.2	10.7	12.4	11.3	12.4	10.4	9.9		
		MCWBR	20.0	19.6	20.1	22.0	21.1	21.8	21.9	20.0	22.3	22.6	21.3	22.5		
(40) Clear-Sky Solar Irradiance	taub		0.378	0.382	0.366	0.342	0.326	0.313	0.315	0.350	0.370	0.386	0.368	0.374		
	taud		2.478	2.473	2.498	2.543	2.559	2.568	2.556	2.449	2.387	2.379	2.463	2.476		
	Ebn at Noon		305	297	289	279	267	263	268	272	283	292	305	307		
	Edh at Noon		37	36	33	28	24	23	24	30	36	39	37	37		
(44) All-Sky Solar Radiation	RadAvg		2260	2017	1682	1278	979	858	927	1184	1562	1877	2175	2343		
	RadStd		106	100	113	63	44	46	58	74	93	130	144	154		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)	+0.68	+1.24	+1.51	N/A	N/A	N/A	N/A	N/A	-153	+256	+81

Nomenclature: See separate page